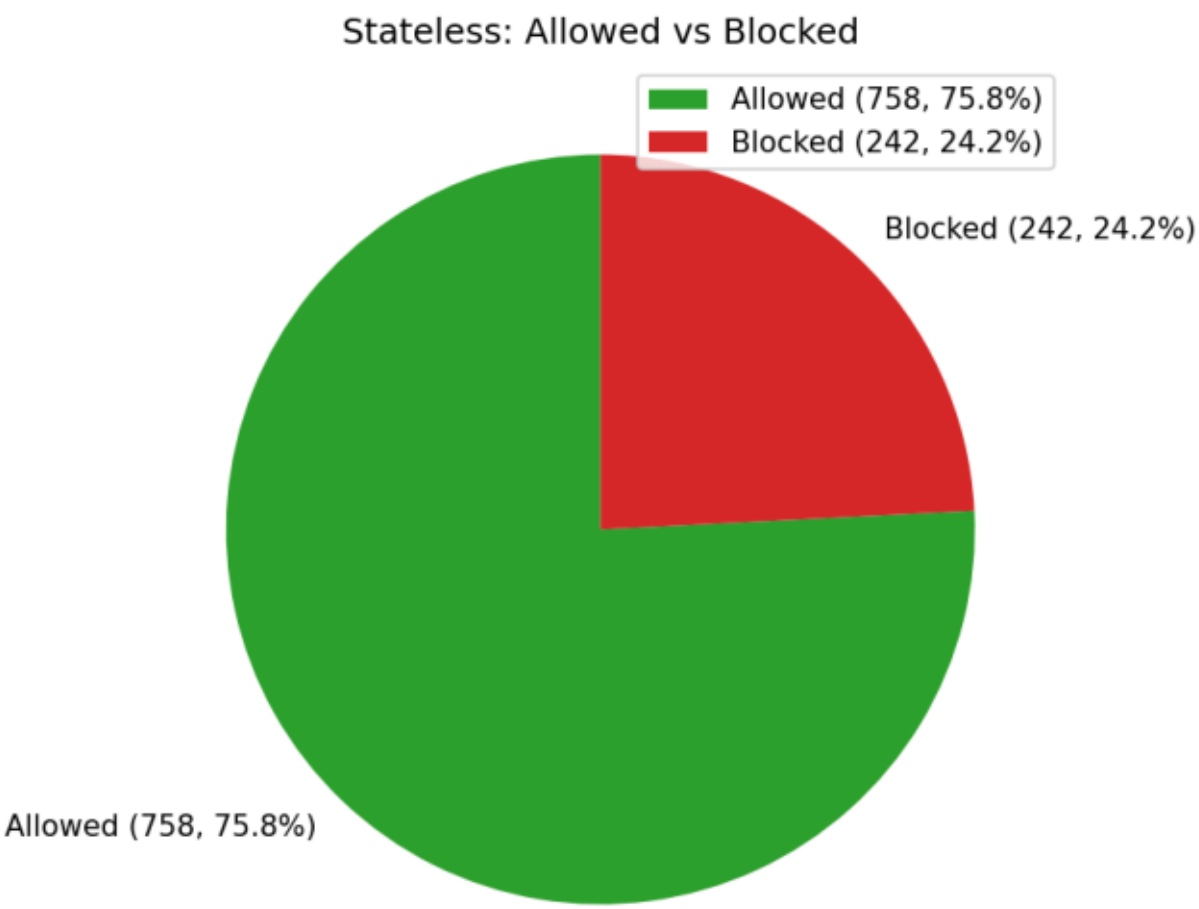
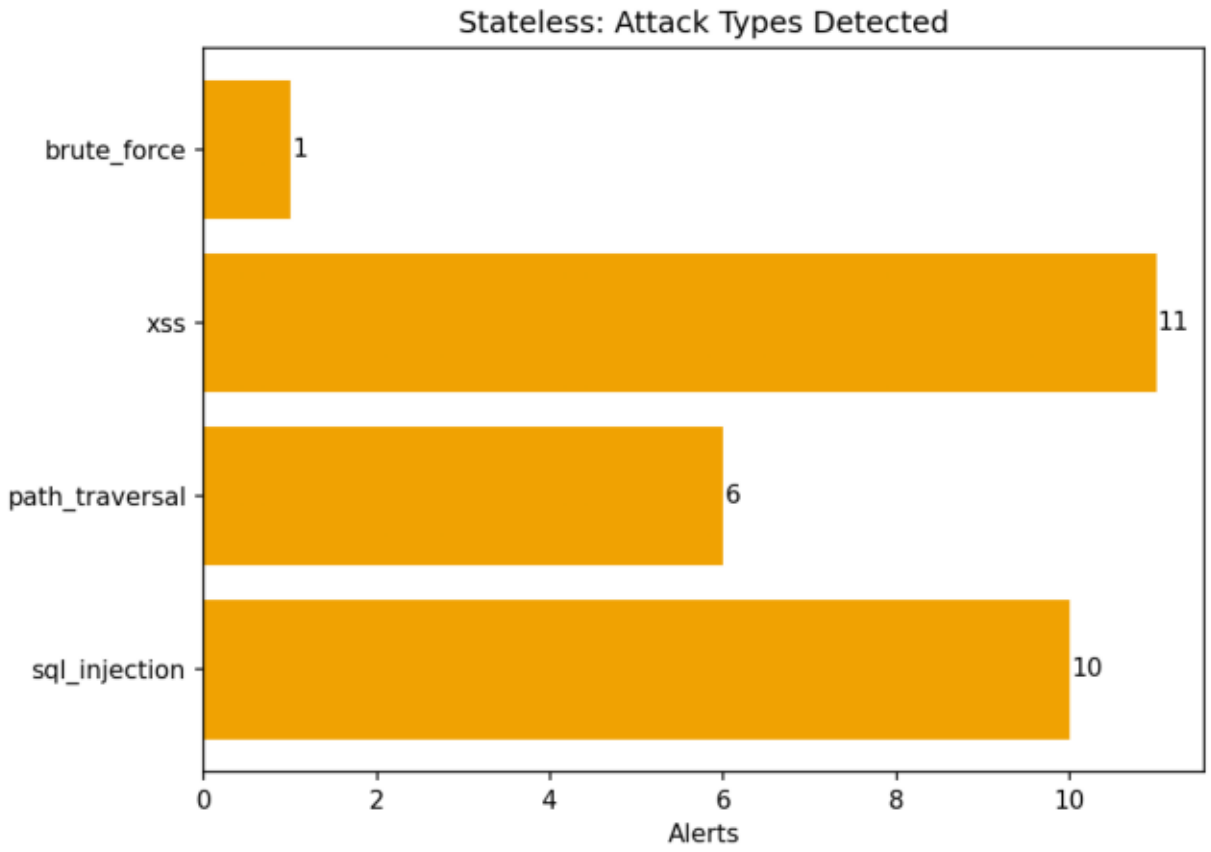
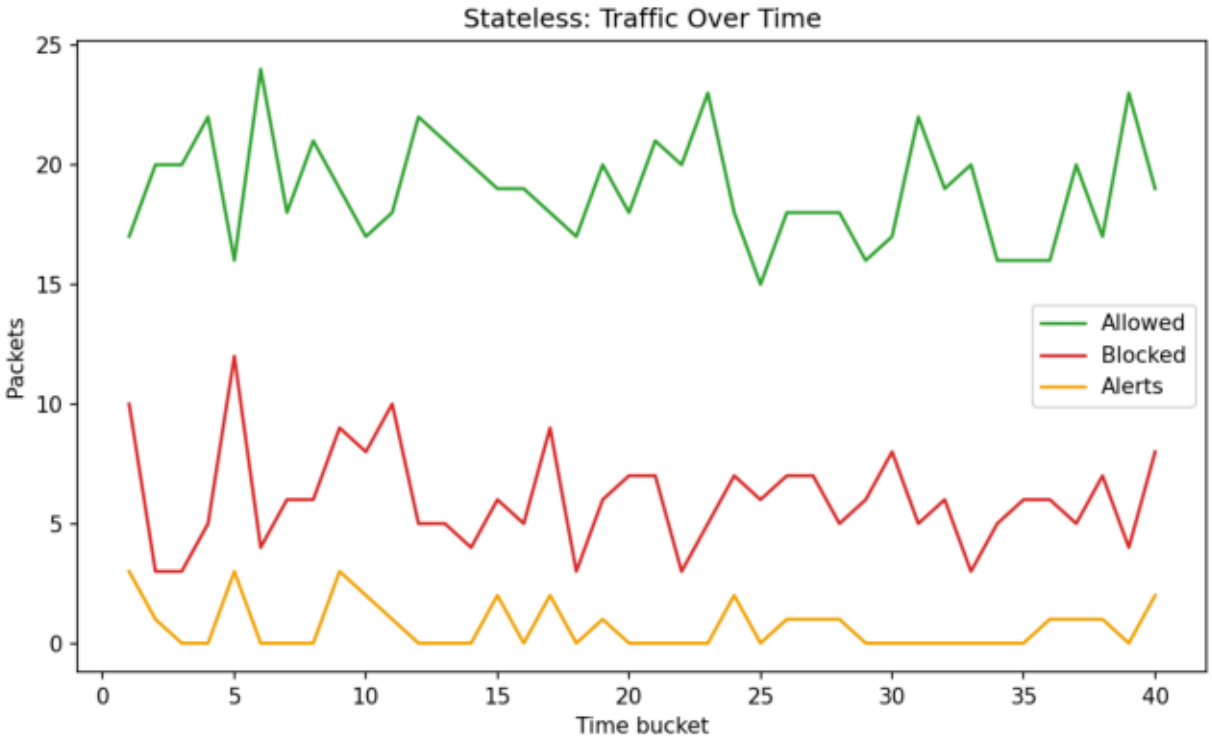
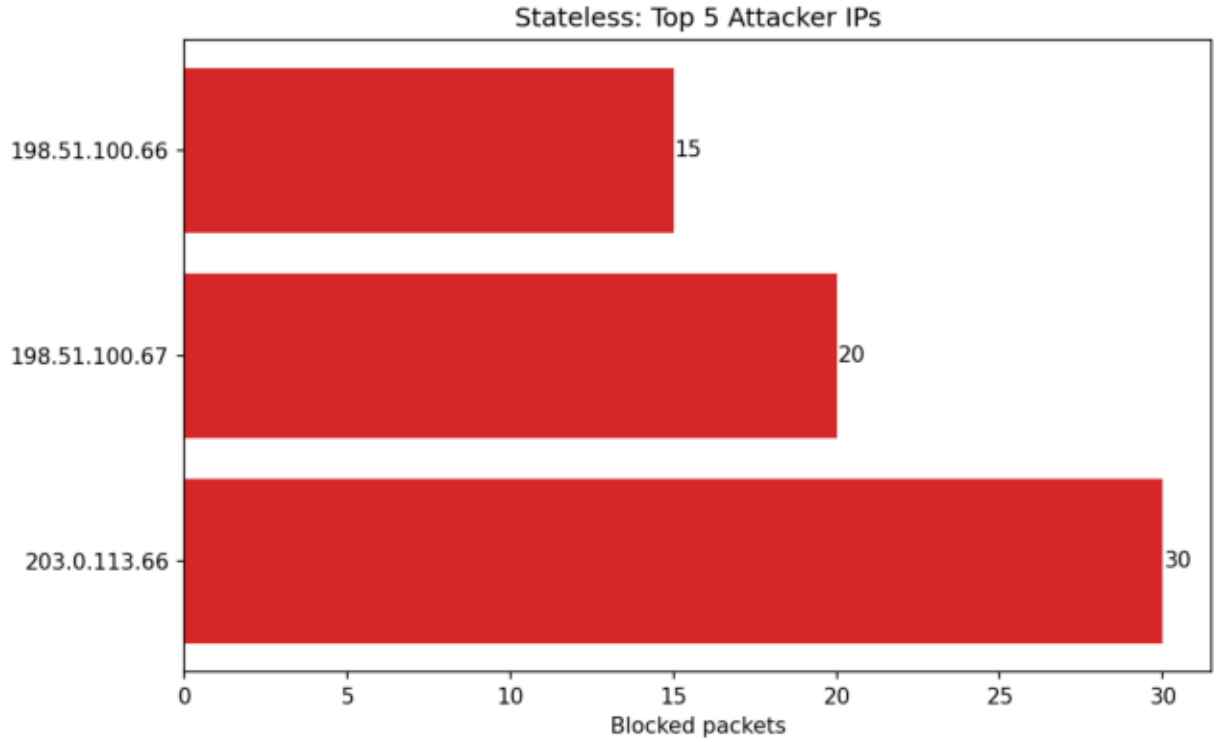
EXECUTIVE SUMMARY
Stateful reduced false positives from 177 to 0 and false negatives from 7 to 4.

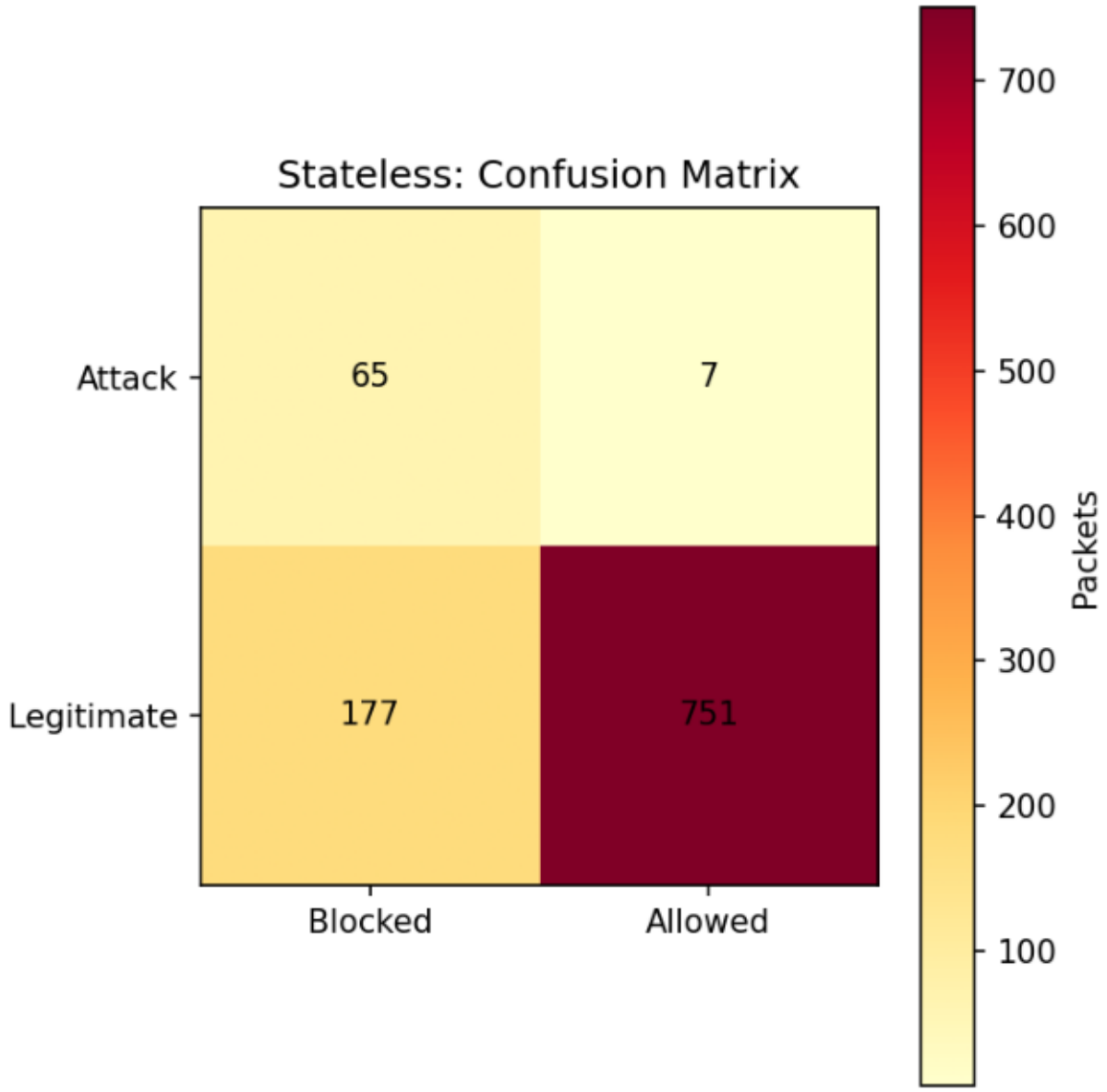
tp: stateless=65, stateful=68
tn: stateless=751, stateful=928
fp: stateless=177, stateful=0
fn: stateless=7, stateful=4
accuracy: stateless=0.816, stateful=0.996
precision: stateless=0.26859504132231404, stateful=1.0
recall: stateless=0.9027777777777778, stateful=0.9444444444444444
f1: stateless=0.41401273885350315, stateful=0.9714285714285714

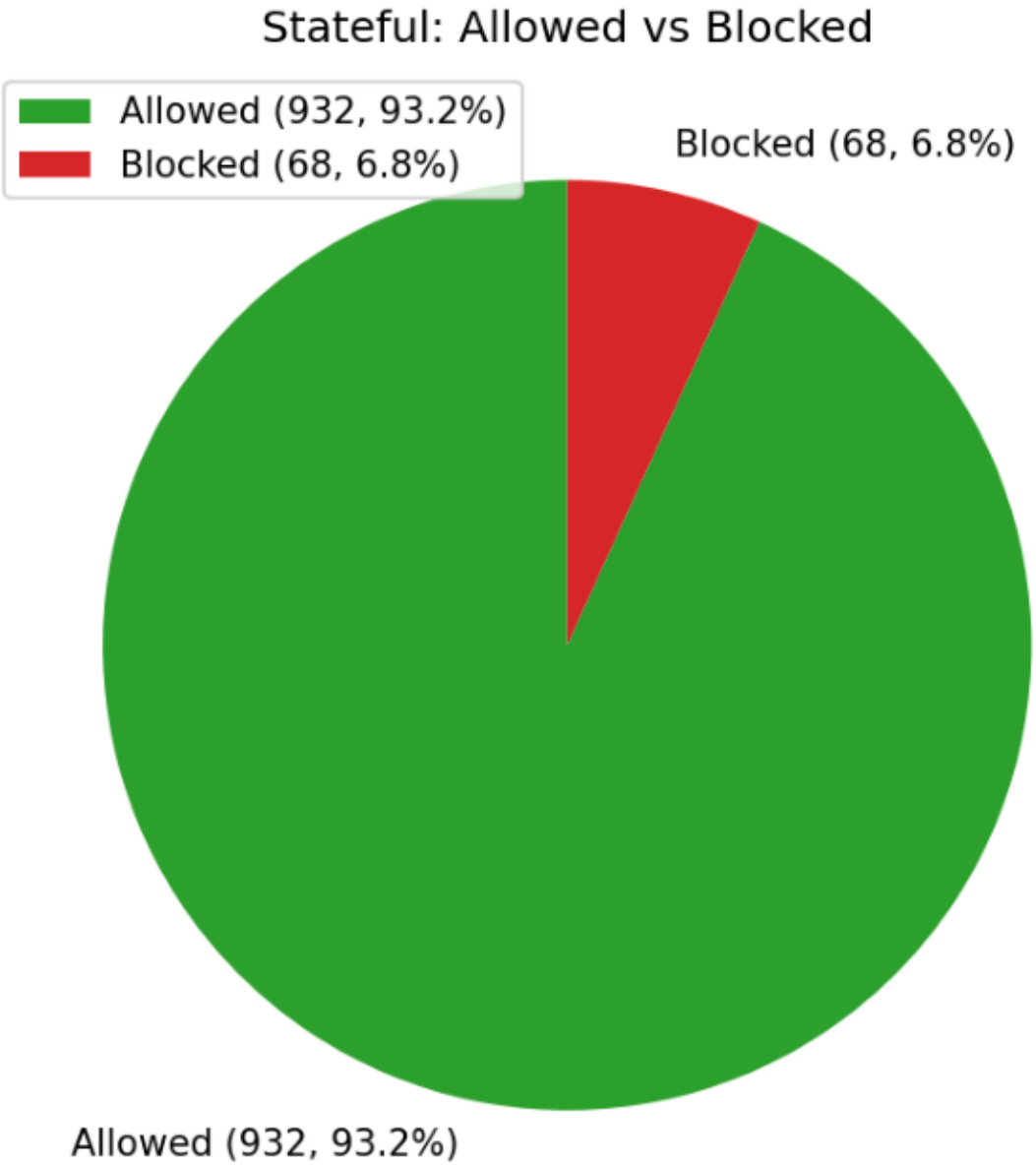

Stateless: Allowed vs Blocked
Allowed (758, 75.8%)
Blocked (242, 24.2%)
Blocked (242, 24.2%)
Allowed (758, 75.8%)

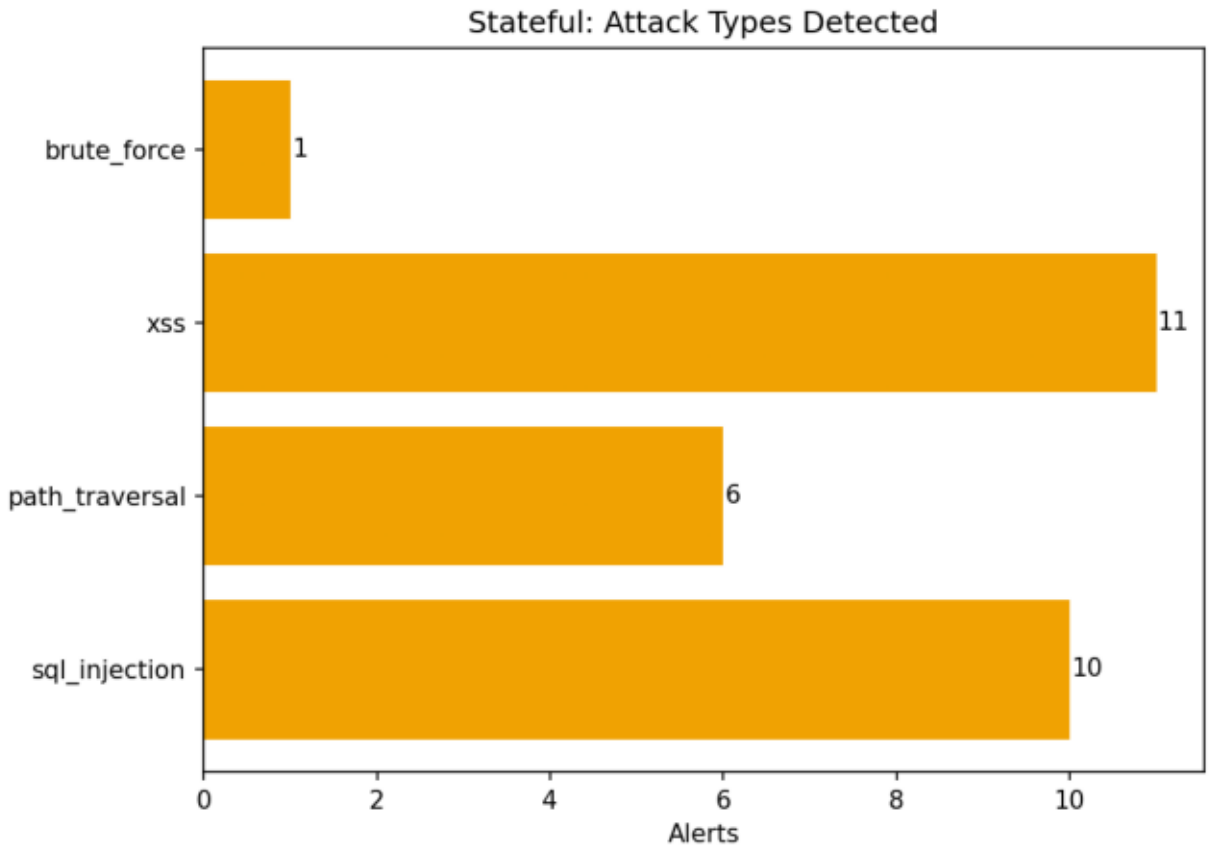

Stateless: Attack Types Detected
brute_force
xss
path_traversal
sql_injection
1
11
6
10
0
2
4
6
8
10
Alerts

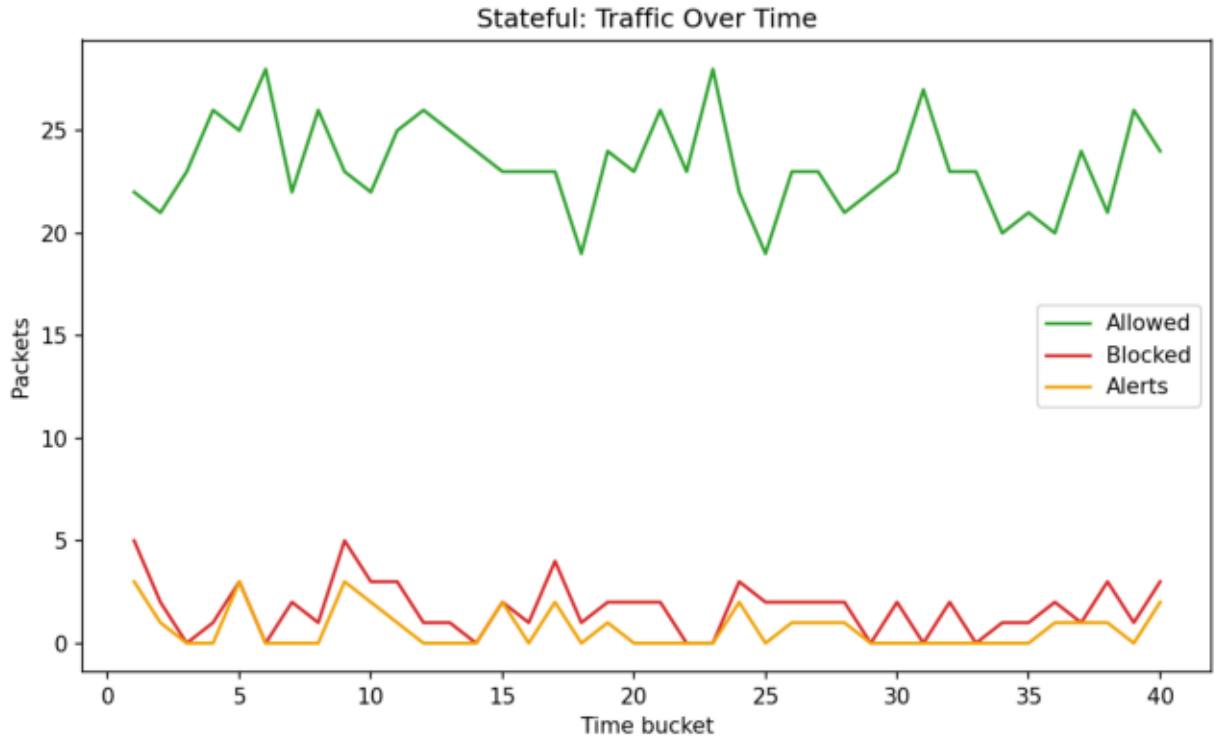

Stateless: Traffic Over Time
Packets
Time bucket
Allowed
Blocked
Alerts
0
5
10
15
20
25
30
35
40

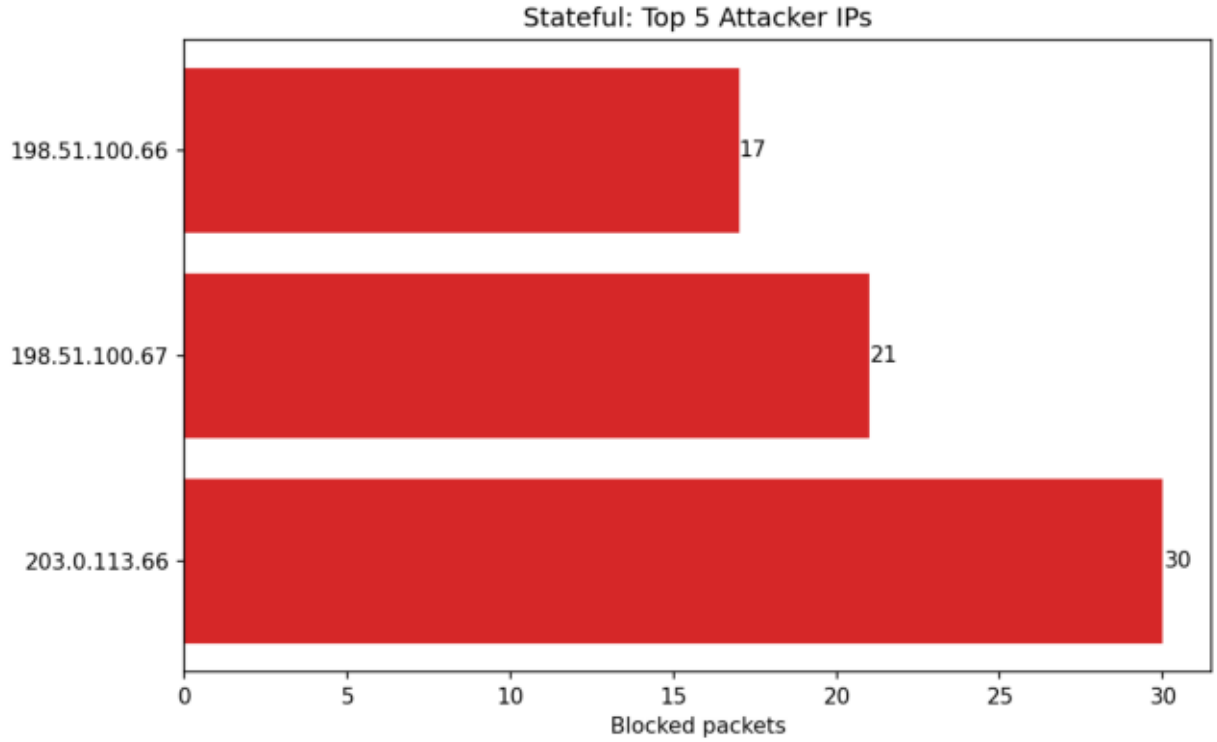

Stateless: Top 5 Attacker IPs
198.51.100.66
198.51.100.67
203.0.113.66
15
20
30
0
5
10
15
20
25
30
Blocked packets

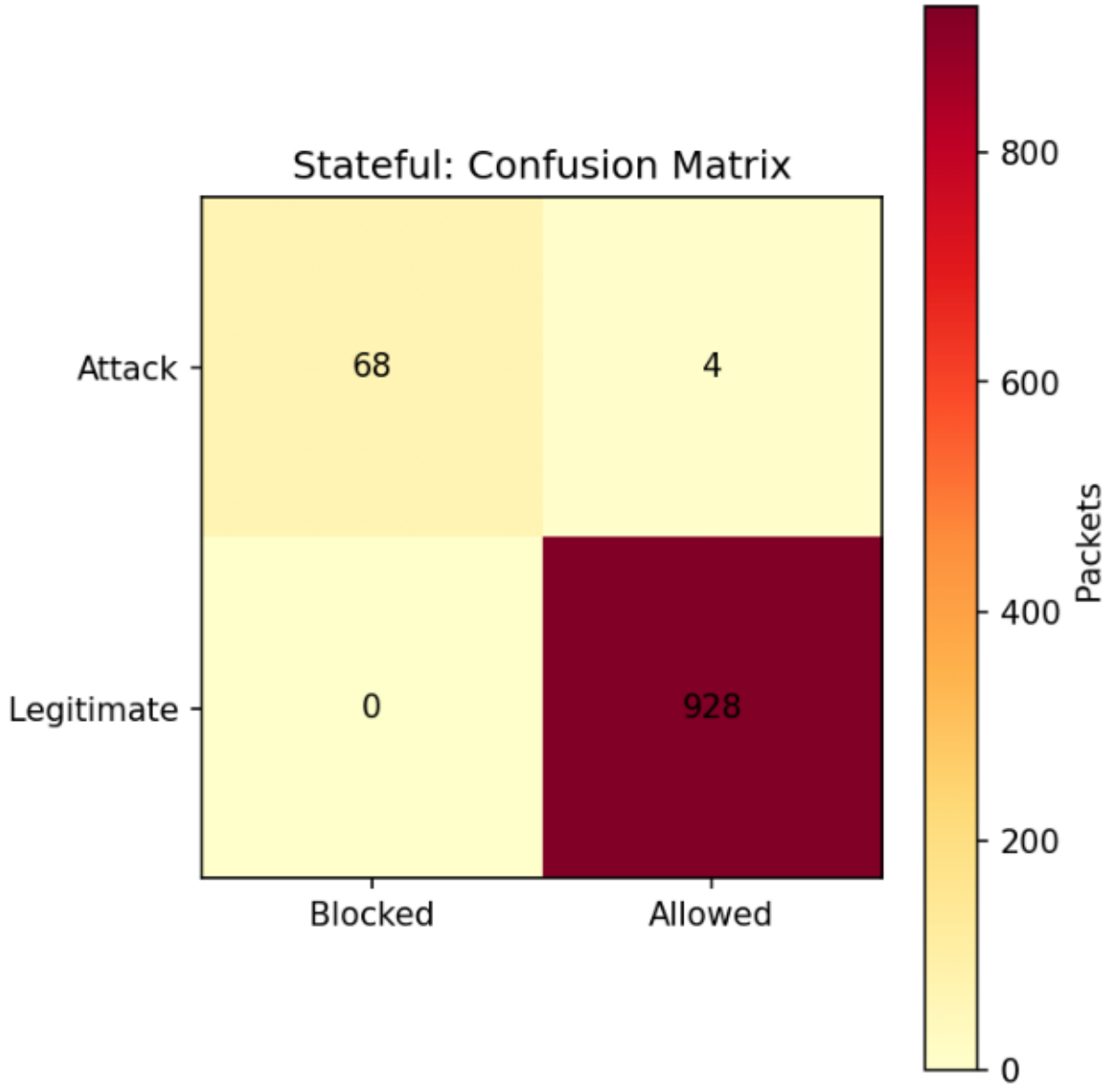

Stateless: Confusion Matrix
Attack
Legitimate
65
7
177
751
Blocked
Allowed
Packets
700
600
500
400
300
200
100

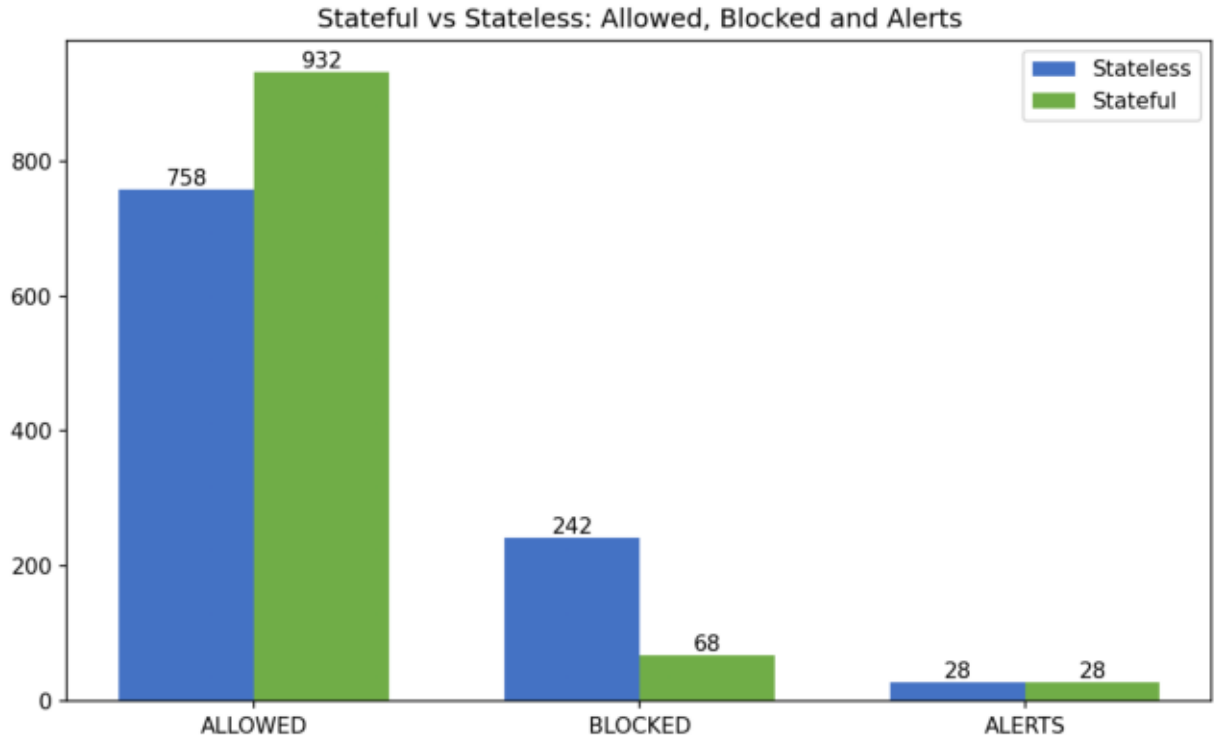

Stateful: Allowed vs Blocked
Allowed (932, 93.2%)
Blocked (68, 6.8%)
Blocked (68, 6.8%)
Allowed (932, 93.2%)

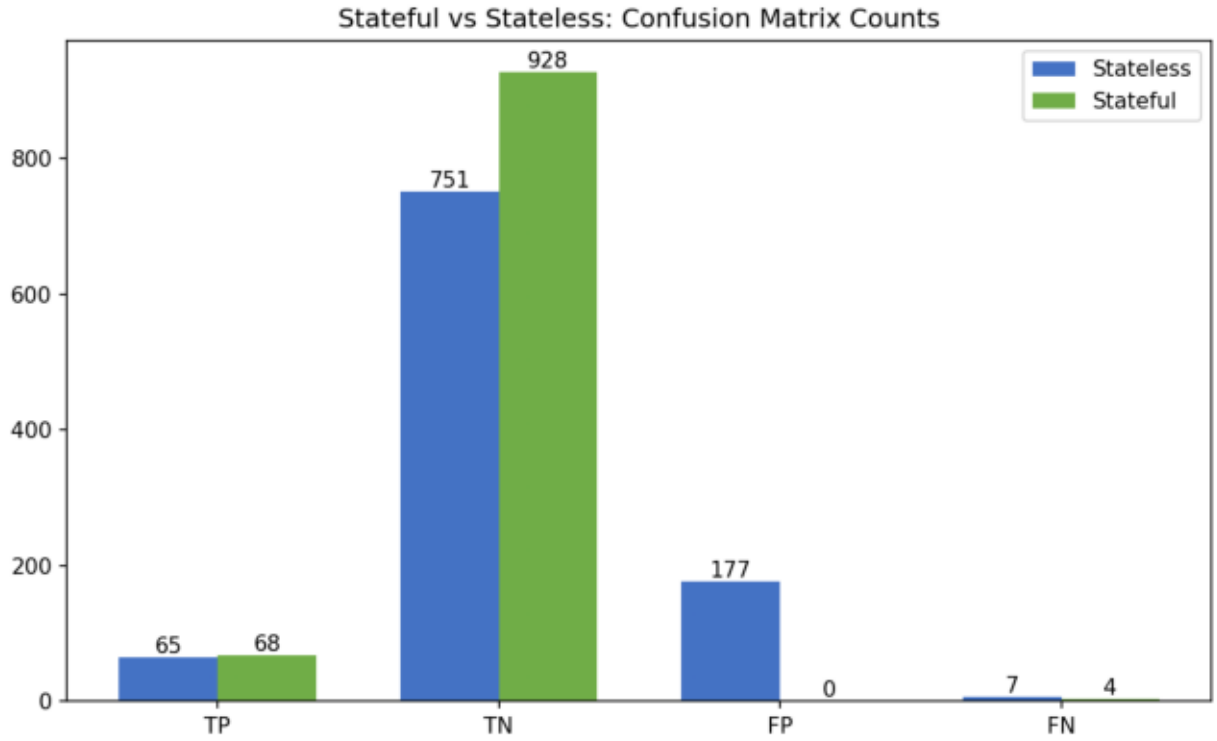

Stateful: Attack Types Detected
brute_force
xss
path_traversal
sql_injection
1
11
6
10
0
2
4
6
8
10
Alerts

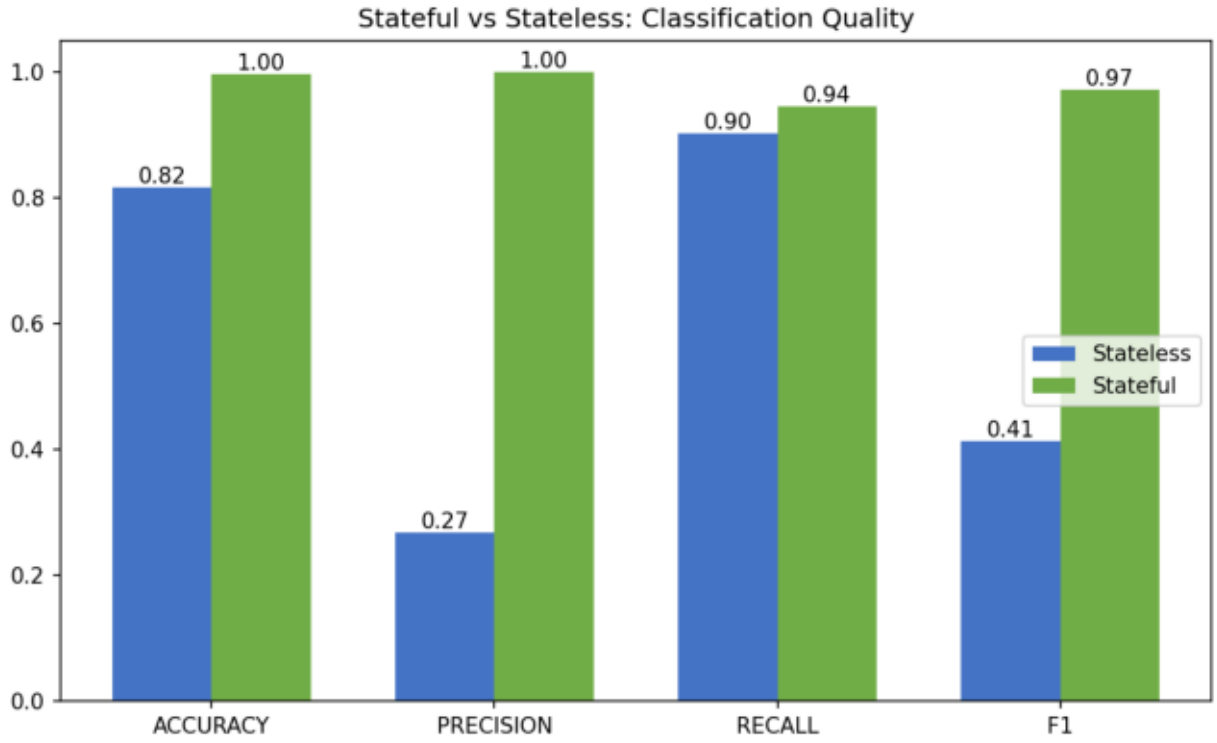

Stateful: Traffic Over Time
Packets
Time bucket
Allowed
Blocked
Alerts
0
5
10
15
20
25
30
35
40


Stateful: Top 5 Attacker IPs
198.51.100.66
198.51.100.67
203.0.113.66
17
21
30
0
5
10
15
20
25
30
Blocked packets


Stateful: Confusion Matrix
Attack
Legitimate
68
4
0
928
Blocked
Allowed
800
600
400
200
0
Packets


Stateful vs Stateless: Allowed, Blocked and Alerts
Stateless
Stateful
800
600
400
200
0
758
932
242
68
28
28
ALLOWED
BLOCKED
ALERTS


Stateful vs Stateless: Confusion Matrix Counts
Stateless
Stateful
800
600
400
200
0
65
68
751
928
177
0
7
4
TP
TN
FP
FN


Stateful vs Stateless: Classification Quality
1.0
0.8
0.6
0.4
0.2
0.0
0.82
1.00
0.27
1.00
0.90
0.94
0.41
0.97
Stateless
Stateful
ACCURACY
PRECISION
RECALL
F1

CONSOLE OUTPUT
Stateful vs Stateless comparison

Mode	Tp	Tn	Fp	Fn	Accuracy	Precision	Recall	F1
stateless	65	751	177	7	0.816	0.269	0.903	0.414
stateful	68	928	0	4	0.996	1.000	0.944	0.971